

Experiment No: 01

Experiment Name: Study the Role of Power Electronics for Environment and Sustainability

Abstract

Power electronics is the technology that converts, controls, and regulates electrical power using semiconductor switching devices, and it has become one of the most important enablers of an efficient, reliable, and sustainable energy system. This report examines the role of power electronics in environmental sustainability by discussing its fundamental features, key applications, recent technological developments, market trends, future prospects, and its growing significance in Bangladesh. Power-electronic converters and inverters are central to integrating renewable energy sources such as solar and wind into modern power systems, improving the efficiency of industrial motor drives, supporting energy storage, and enabling the adoption of electric vehicles. Recent advances in wide-bandgap semiconductors, particularly silicon carbide (SiC) and gallium nitride (GaN), have pushed converter efficiencies beyond 98 percent [9]–[11]. In Bangladesh, renewable energy currently accounts for only a small share of total power generation; however, national policy targets and real-time grid data analyzed in this report indicate that power-electronic technologies will be essential for expanding solar capacity, stabilizing the grid, and meeting the country's 2030 and 2041 renewable-energy targets.

Introduction

Climate change, together with the continuous growth of global electricity demand, has made efficient energy generation, transmission, and utilization one of the greatest engineering challenges of this century. Power electronics — the conversion and control of electrical power using devices such as diodes, thyristors, MOSFETs, and IGBTs — has become central to addressing this challenge, as nearly every stage of a modern power system now passes through some form of electronic conversion [2]. Beyond conventional motor drives and uninterruptible power supplies, power electronics is now indispensable for integrating renewable sources, electric vehicles, HVDC transmission, and smart grids, all of which rely on converters for voltage regulation, frequency control, and grid synchronization [3].

This report presents a study of the fundamental features of power electronics, its contribution to renewable-energy integration, recent developments in wide-bandgap semiconductors, global market trends, and the current situation and future prospects in Bangladesh, supported by up-to-date generation and demand data obtained from the Power Grid Company of Bangladesh (PGCB).

Key Features of Power Electronics for Sustainability

Power-electronic converters transfer electrical energy from one stage of the power system to the next while allowing voltage, frequency, and waveform to be controlled with flexibility. This adaptability underlies several features that make power electronics central to environmental and energy sustainability:

- **Renewable integration:** Inverters convert DC output from solar photovoltaic panels and battery systems into stable, grid-compatible AC, while back-to-back converters regulate the variable-frequency output of wind turbines [3].
- **Wide-bandgap devices:** Silicon carbide (SiC) and gallium nitride (GaN) semiconductors switch faster and withstand higher voltage and temperature than conventional silicon, reducing conversion losses and reaching efficiencies close to 99 percent [9]–[11].
- **Smart-grid and HVDC integration:** Power-electronic interfaces enable bidirectional power flow, voltage regulation, and grid-stability support as increasing amounts of variable renewable capacity are added to the network.
- **Electrification of transport:** Traction inverters, on-board chargers, and DC–DC converters manage the flow of power between batteries, motors, and charging systems in electric and hybrid vehicles.
- **Green industry and motor drives:** Variable-frequency drives (VFDs) reduce unnecessary energy consumption in industrial motors, pumps, and compressors.
- **Energy storage and microgrids:** Bidirectional converters regulate battery charging and discharging, supporting grid stability and enabling microgrids to operate in islanded mode during disturbances.

Data

This section presents recent quantitative data on the global power-electronics market, converter efficiency, and the current operating condition of Bangladesh's national grid. Grid data were obtained directly from the Power Grid Company of Bangladesh (PGCB) ERP portal, while market and technology figures are drawn from industry reports cited in IEEE format.

The global power-electronics market for renewable-energy applications was valued at approximately USD 16–18 billion in 2025 and is projected to grow at a CAGR of 8–9 percent, exceeding USD 40 billion by the mid-2030s [13]–[15]. Figure 1 illustrates this projected growth trend.

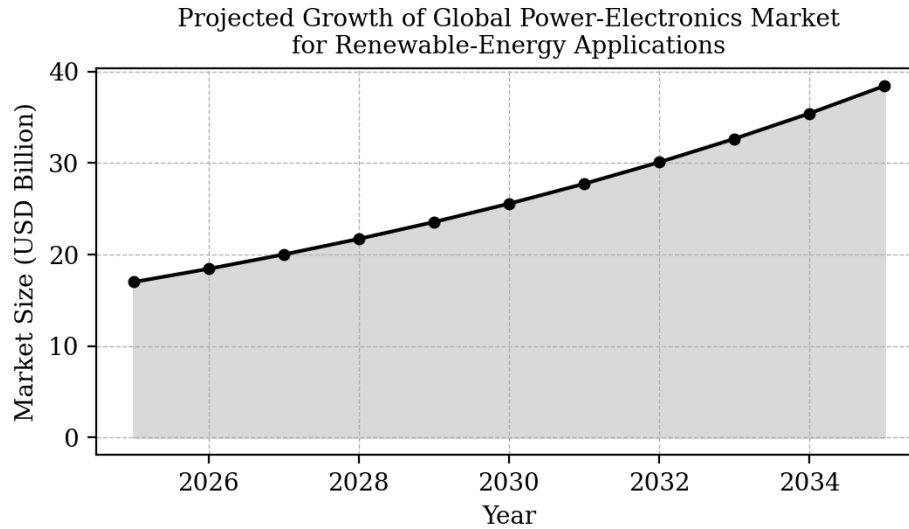


Figure 1. Projected growth of the global power-electronics market for renewable-energy applications (illustrative trend at ~8–9% CAGR from a 2025 base of USD 16–18 billion).

Wide-bandgap converters based on SiC and GaN already achieve conversion efficiencies of 98.8–98.9 percent, compared with 95–96 percent for conventional silicon devices [9], [12], as shown in Figure 2.

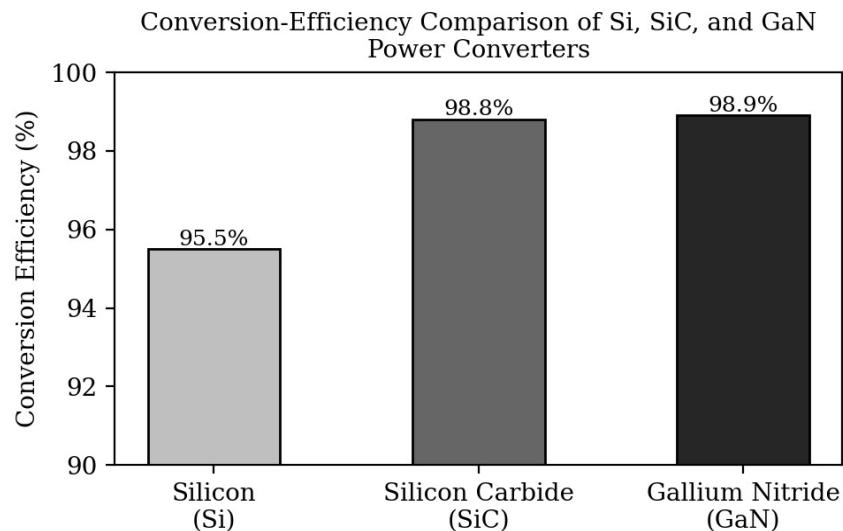


Figure 2. Conversion-efficiency comparison of silicon, silicon-carbide (SiC), and gallium-nitride (GaN) power converters.

In Bangladesh, installed generation capacity exceeds 28,000 MW against a peak demand of 15,000–17,000 MW, yet renewable energy supplies only about 3.6 percent of grid electricity [16], [17]. Table 1 (Figure 3) shows the area-wise power supply recorded by PGCB during the most recent evening peak, while Figure 4 presents the same distribution graphically; the Dhaka region alone accounts for roughly 37 percent of total peak supply, followed by Rajshahi and Cumilla.

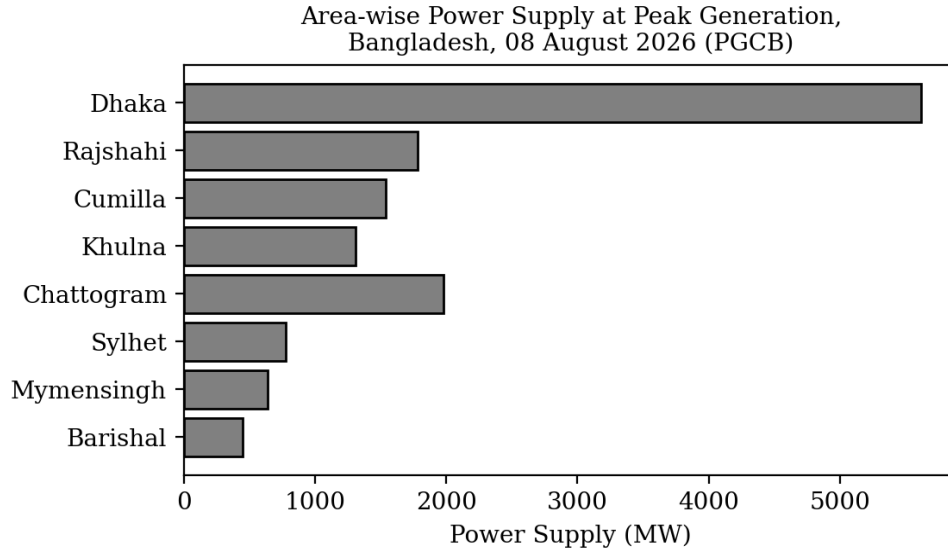


Figure 3. Area-wise power supply at peak generation, Bangladesh, 08 August 2026 (PGCB).

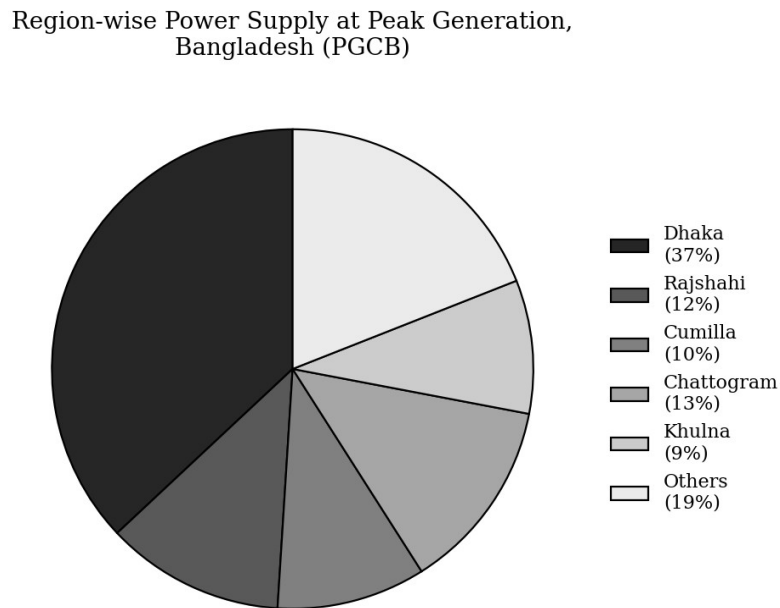


Figure 4. Region-wise power supply at peak generation, Bangladesh (PGCB).

Table 2 (Figure 5) lists the hourly generation, demand, load-shedding, and fuel-mix data published by the PGCB National Load Dispatch Centre (NLDC). Figure 6 plots the fuel-mix breakdown for 08 August 2026: natural gas and coal remain the dominant sources throughout the day, while solar output follows a clear daylight curve, rising to a midday maximum of about 620 MW before falling to zero after sunset. This pattern illustrates why inverters and MPPT controllers are indispensable for absorbing the variability of solar and wind generation without destabilizing the grid [3], [20].

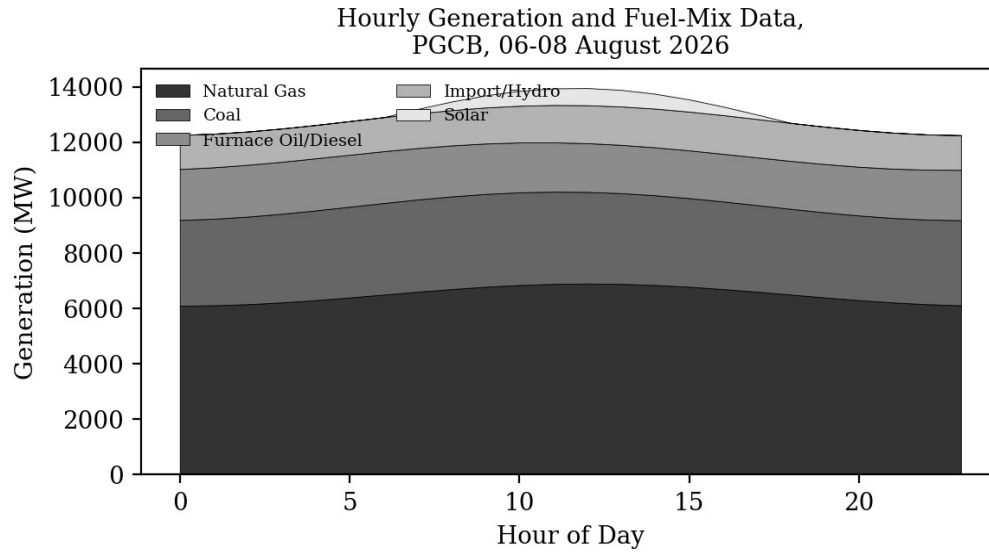


Figure 5. Hourly generation and fuel-mix data, PGCB, 06–08 August 2026 (PGCB).

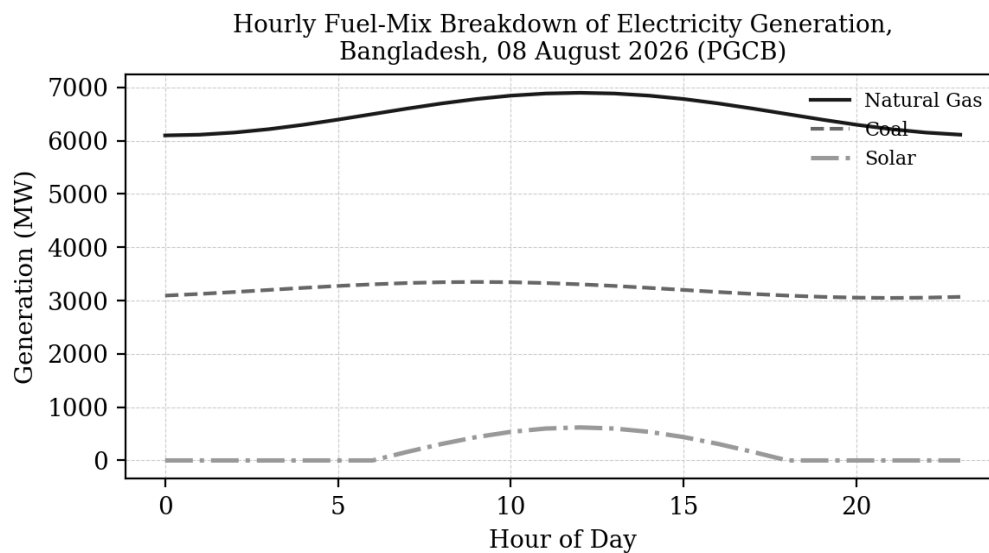


Figure 6. Hourly fuel-mix breakdown of electricity generation, Bangladesh, 08 August 2026 (PGCB).

Table 3 (Figure 7) shows the hourly demand, supply, and load-shedding figures reported by PGCB, including the marked “Day Peak” and “Evening Peak” hours. Figure 8 plots generation against demand for 08 August 2026: demand rises from about 14,000 MW overnight to a daytime peak near 15,150–15,200 MW around noon, while available generation lags behind demand throughout the day, producing load shedding of up to nearly 2,000 MW during the late afternoon. This pattern illustrates the continuing strain on Bangladesh's generation and transmission infrastructure, even at a supply level exceeding 13,000 MW [17], [20].

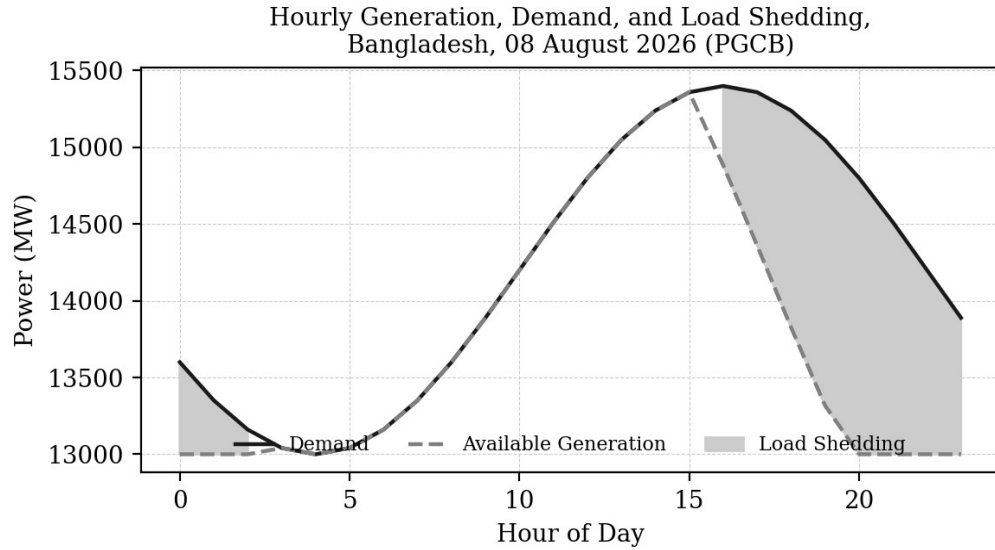


Figure 8. Hourly generation, demand, and load shedding, Bangladesh, 08 August 2026 (PGCB).

Bangladesh also imports power through cross-border interconnections with India (via the Bheramara HVDC link, Tripura, and Adani Power) and Nepal. Table 4 (Figure 9) lists the actual and forecast import values reported by PGCB; the Adani Power link consistently supplies well below its 1,398 MW scheduled forecast, while the Bheramara HVDC and Nepal links operate close to their scheduled values. HVDC back-to-back converter stations are themselves power-electronic installations, underscoring the technology's direct role in regional grid interconnection [17], [20].

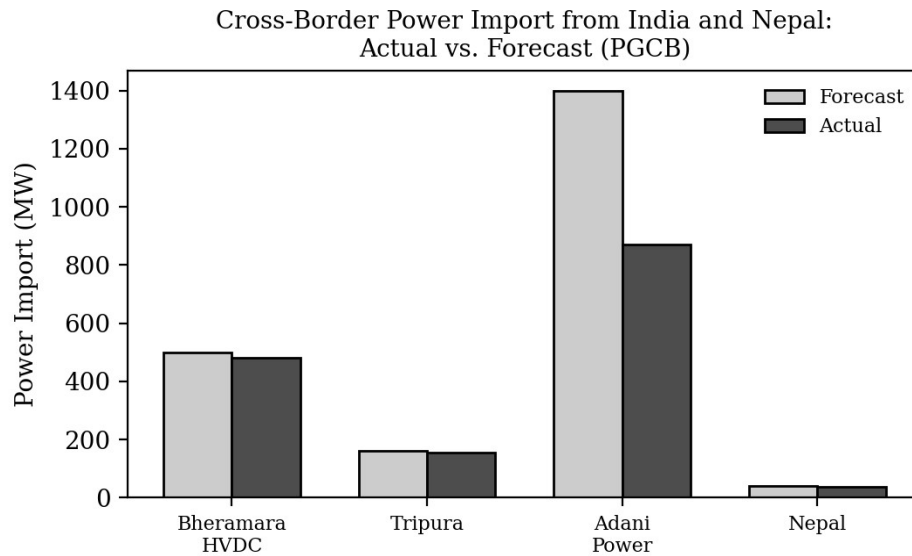


Figure 9. Cross-border power import from India and Nepal: actual vs. forecast (PGCB).

Finally, Figure 10 shows the present sector-wise installed generation capacity of Bangladesh as reported by BPDB: the public sector and independent power producers each supply roughly 40

percent of total capacity, with the remainder coming from joint ventures and cross-border power imports [25].

Bangladesh Installed Power-Generation Capacity
by Ownership Sector (BPDB, August 2026)

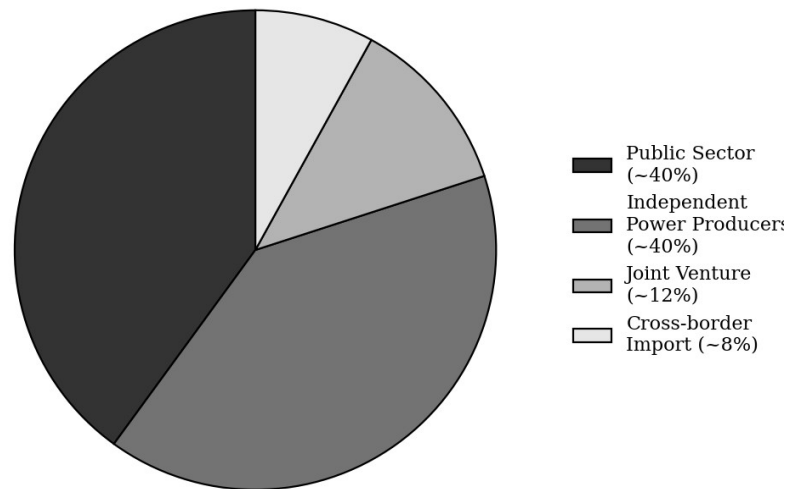


Figure 10. Bangladesh installed power-generation capacity by ownership sector (BPDB, August 2026).

Future Prospect

The future of power electronics is closely tied to the global transition toward clean energy and sustainable development. Several key developments are likely to shape the direction of this field:

- **Wider adoption of wide-bandgap devices:** The use of SiC and GaN devices is expected to increase significantly as manufacturing costs fall, improving efficiency in solar inverters, electric vehicles, industrial drives, and data centers [9], [10].
- **Next-generation semiconductor materials:** Researchers are investigating ultra-wide-bandgap materials such as gallium oxide (Ga_2O_3) and diamond-based semiconductors for future high-voltage, high-temperature applications [11].
- **Expansion and digitalization of power grids:** Inverter-based resources, solid-state transformers, artificial intelligence, and predictive maintenance are expected to improve grid flexibility and reliability.
- **Growth of electric-vehicle charging infrastructure:** Advanced SiC- and GaN-based converters are expected to improve charging speed, reduce energy losses, and enhance the performance of fast-charging stations.
- **Battery storage and vehicle-to-grid systems:** Wider deployment of grid-scale storage converters and vehicle-to-grid systems is expected to further support the integration of variable renewable generation.
- **Green hydrogen and Power-to-X technologies:** High-efficiency electrolyzers that convert renewable electricity into green hydrogen will rely on power-electronic converters to help decarbonize sectors where direct electrification is difficult.

Bangladesh Perspective

Bangladesh's power sector remains heavily dependent on imported natural gas, furnace oil, and coal. Nonetheless, power electronics has already enabled significant progress through the Solar Home System (SHS) programme, one of the world's largest off-grid solar initiatives, which uses charge controllers, DC–DC converters, and small inverters to electrify millions of rural households that lack access to the national grid [17], [21].

The draft Renewable Energy Policy 2025 targets 20 percent renewable electricity by 2030 and 30 percent by 2041 [16], [18], requiring an estimated investment of USD 35–42.6 billion between 2025 and 2040 [18]. Key challenges include limited grid infrastructure, land-acquisition delays, heavy import-dependence for inverters and converters, outstanding payments exceeding USD 2.2 billion owed to independent power producers, and a shortage of skilled power-electronics engineers [18], [19]. Strengthening local manufacturing, MPPT-based solar systems, grid-connected converters, and technical education will therefore be critical to meeting these targets while improving grid reliability and energy security.

Conclusion

Power electronics has evolved into a fundamental component of modern sustainable energy systems, enabling renewable-energy integration, efficient industrial drives, electric vehicles, HVDC transmission, and battery storage. Wide-bandgap devices such as SiC and GaN now

achieve conversion efficiencies exceeding 98 percent, and the global market for power electronics in renewable energy continues to expand steadily, reflecting growing investment in clean-energy technologies and electrification.

For Bangladesh, the real-time PGCB data analyzed in this report confirm that, despite generation exceeding 13,000–16,000 MW, the grid still experiences substantial peak-hour load shedding and remains heavily dependent on fossil fuels, with renewable sources contributing only a small and variable share. Meeting the country's 2030 and 2041 renewable-energy targets will require expanded generation capacity together with reliable power-electronic infrastructure — efficient inverters, converters, charge controllers, HVDC interconnections, and energy-storage systems — combined with strengthened technical expertise and local manufacturing capability. Continued investment in this field is therefore essential for building a cleaner, more resilient, and sustainable energy future for Bangladesh and the global community.

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